# Anxin (Bob) Guo

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## Education

**Northwestern University**, Evanston, Illinois — September 2023 – Present
Ph.D. student in Computer Science. Cumulative GPA: 3.867/4.000.

**Northwestern University**, Evanston, Illinois — September 2019 – June 2023
B.A. in Mathematics and M.S. in Computer Science. Undergraduate cumulative GPA: 3.821/4.000.

## Research Interests

Theoretical computer science and the foundations of machine learning, with interests in the capabilities and limitations of learning and computation, and how language models learn, memorize, and reason.

## Research Projects and Papers

### Manuscripts and Ongoing Projects

**Approximate Syzygies of Smoothed Polynomials and Applications**
Aditya Bhaskara, **Anxin Guo**, Tanmay Sinha, Vaidehi Srinivas, and Aravindan Vijayaraghavan.
*Manuscript.* PDF.

**Learning Rules, Memorizing Facts: An Information-Theoretic View of Hallucination**
Gengze Xu, **Anxin Guo**, Yifei Hu, Xiangyu Wu, Yong Liu, and Ziqiao Wang.
*Preprint, 2026.* arXiv.

**How Much Do You Trust A Dog? Quantifying Hint-Following in Reasoning Language Models**
**Anxin Guo**, Chirag Pabbaraju, Sara Ahmadian, Kamesh Munagala, and Aravindan Vijayaraghavan.
*Preprint, 2026.* arXiv.

**Regularized Transport Weights: a Principled Approach for Active Domain Adaptation**
**Anxin Guo**, Tanmay Sinha, Dionysis Arvanitakis, Vaidehi Srinivas, Alex Drlica-Wagner, Alexander Ji, and Aravindan Vijayaraghavan.
*Manuscript.*

**Leanify: Verified Proof Compression and Reusable Lemma Discovery**
Ruize Xu, **Anxin Guo**, Yi Gu, Tianchong Jiang, Chenxiao Yang, Tianhao Wang, and Zhiyuan Li.
*Manuscript.*

**Copying Under Repetition: From Prefix Matching to Position Hashing**
**Anxin Guo**, Tianlong Huang, Chenxiao Yang, and Zhiyuan Li.
*Preprint, 2026.* arXiv.

## Publications

**Hallucination is a Consequence of Space-Optimality: A Rate-Distortion Theorem for Membership Testing**
**Anxin Guo** and Jingwei Li.
*International Conference on Machine Learning (ICML), 2026, spotlight, top 2.2%.* arXiv.

**Agnostic Learning of Arbitrary ReLU Activation under Gaussian Marginals**
**Anxin Guo** and Aravindan Vijayaraghavan.
*Conference on Learning Theory (COLT), 2025.* arXiv; conference version.

**To Store or Not to Store: A Graph-Theoretic Approach for Dataset Versioning**
**Anxin Guo**, Jingwei Li, Pattara Sukprasert, Samir Khuller, Amol Deshpande, and Koyel Mukherjee.
*IEEE International Parallel and Distributed Processing Symposium (IPDPS), 2024.* arXiv; conference version.

## Selected Talks

**Simons Institute** — Student spotlight — October 7, 2026
*Hallucination is a Consequence of Space-Optimality.*
Workshop: *Trustworthy AI: From Hallucinations to Reliable Autonomy.*

**SkAI Works-In-Progress** — Joint presentation — September 30, 2026
*Intelligent Scheduling for Astronomical Surveys.*
With Rachel Hur and Tanmay Sinha.

**Google AI Study Group** — Invited talk — September 14, 2026
*Why LLM Hallucinate? A Perspective from Space-Optimality.*

**IDEAL Annual Meeting, TTIC** — Lightning talk — September 10, 2026
*Hallucination is a Consequence of Space-Optimality.*

**IDEAL Annual Meeting, UIC** — Student talk — June 9, 2025
*Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.*

**Northwestern University** — CS Theory Seminar — February 6, 2025
*Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.*

**NSF TRIPODS Workshop, TTIC** — Lightning talk — December 7, 2024
*Agnostic Learning of Arbitrary ReLU Neuron under Gaussian Marginals.*

## Awards and Honors

**Ph.D. Student Research Award**, Computer Science Department, Northwestern University — 2024 – 2025
**Barris Award for Outstanding TA**, CS 335: Introduction to the Theory of Computing — Fall 2024
**Junior Career Award in Mathematics**, Mathematics Department, Northwestern University — 2021 – 2022

## Research and Teaching

**IDEAL-funded summer research exchange**, University of Chicago — Summer 2026
Hosted by Chao Gao.

**Research Intern / IDEAL-funded summer research exchange**, TTIC — Summer 2025
Toyota Technological Institute at Chicago. Hosted by Zhiyuan Li.

**Teaching Assistant**, CS 336: Design and Analysis of Algorithms — Spring 2026

**Teaching Assistant**, CS 335: Introduction to the Theory of Computing Fall 2024

- Gave a guest lecture on busy beaver numbers in addition to regular TA duties.
- Awarded the Barris Award for Outstanding TA.

## Service and Programs

**Program Committee / Reviewer**, Reliable ML from Unreliable Data, NeurIPS Workshop 2025
**Volunteer**, 65th IEEE Symposium on Foundations of Computer Science (FOCS) 2024
**Ross Mathematics Program (Asia)**, student, junior counselor, counselor 2018 – 2020
**Directed Reading Program**, Linear Representations of Finite Groups Spring 2021
Read with Wenyuan Li.

## Skills

**Technology:** C++, Python (NumPy, PyTorch, pandas, matplotlib), Mathematica, and Gurobi Optimizer.